

Heat Transfer to a Fluid in a Stirred Tank

Final Lab Report #1

CHE 382 - Unit Operations Lab II
February 05, 2020

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1. Executive Summary

The objective of this experiment was to determine the overall heat transfer coefficient and observe its dependence on the impeller speed, temperature, and whether the tank is baffled or unbaffled. We also compared different methods of obtaining the overall heat transfer coefficient to compare their accuracy.

Using the first method, we obtained overall heat transfer coefficient (U) values of 417, 300, and $339 \frac{W}{m^2s}$ with increasing impeller speeds for an unbaffled tank. As a function of increasing temperature, the values of U were 218 and $399 \frac{W}{m^2s}$. For the baffled tank, we obtained U values of 1446, 1940, and $2331 \frac{W}{m^2s}$ for increasing impeller speeds. With increasing temperature, the values of U were 455 and $276 \frac{W}{m^2s}$. Using the second method, the values of U for the unbaffled tank were 843, 860, and $866 \frac{W}{m^2s}$ at increasing impeller speeds. With increasing temperature, the values of U were 477 and $445 \frac{W}{m^2s}$. For the baffled tank, at increasing impeller speeds the measured values of U were 677, 678, and $687 \frac{W}{m^2s}$. At increasing temperatures, the values were 509 and $433 \frac{W}{m^2s}$.

From doing this lab, we observe that the heat transfer coefficient increases with increasing impeller speeds. This is due to the fact that the Reynolds number increases with increasing impeller speeds. The heat transfer coefficient also increases with increasing temperature, assuming the temperature of the fluid inside the vessel remains constant. The presence of baffles enhances mixing which, in turn, increases the Reynolds number and the heat transfer between the surfaces. This yields a higher heat transfer coefficient.

2. Introduction

The aim of this laboratory experiment is to model the heat transfer and other necessary parameters associated with an agitated tank. Also, being able to understand the peculiarities of the heat transfer coefficient between the liquid, the vessel wall, and steam. To chemical engineers around the world, understanding the heat transfer taking place across an equipment is very important as it helps us know for certain whether an equipment of interest can operate safely and ensuring that right materials are deployed in the right places. With this knowledge, coming up with the right countermeasures such as having a cooling system in place to help accommodate the heat generated from an exothermic reaction happening in a reaction vessel becomes achievable. In the industry today, modelling heat transfer is very crucial in order to be able to set appropriate operating conditions and equipment limits while ensuring a safe working environment for workers, profitability and judicious use of resources. [1]

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The experiment set-up includes a jacketed vessel containing inlets for both water and steam, outlets to a condenser and a heat exchanger and a discharge valve at the bottom of the vessel as well as a connection to a pump. There are several thermostats located at every inlet and outlet all around the equipment set-up and a propeller from an impeller used to agitate the fluid in the vessel. For this experiment, the heat transfer is influenced by parameters such as the impeller speed, fluid (water) properties and the presence or absence of baffles. These parameters are changed (and /or varied) to develop a correlation for their interaction with the heat transfer.

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In this experiment, the first part was carried out at steady state and the vessel is subjected to impeller speed changes while keeping flow-rate constant, followed by changing flowrate at constant impeller speeds. The other half is then conducted, basically performing the same experiment as the first half but this time in the presence of baffles, which is deployed to enhance an effective heat transfer. This concept is often encountered in batch manufacturing, where it is necessary to ascertain the time it takes to heat or cool a batch process and or determine the cooling capabilities required to conduct an exothermic or endothermic reaction at a given temperature. [2]

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3. Theory

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Heat transfer through the apparatus requires the understanding of multiple mediums heat transfer. That is, looking at the heat transfer through the liquid, the walls of the vessel, and lastly, to the heat exchanger. At steady state, the condition of the experiment under which this experiment was conducted, it was discovered that the amount of heat transferred, Q , is the same at any point in the outward direction. Also, at steady state there is also no build-up of heat as the heat-flux across the medium(s) are consistent. [3] Fourier's Law is used to describe heat transport through solid surfaces. The use of agitation should improve the heat transfer coefficient for the fluid in the vessel. [4] Using the data collected, calculations for the heat transfer coefficient U are made. The enthalpy values will need to be found to solve for U ; these values will be compared to those from the correlation.

The final goal is to find the H_o (the outer heat transfer) and H_i (the internal heat transfer). We can obtain these key values by measuring the temperature across three distinct points. The first is the inlet/outlet temperature of the cooling coils, the second is the internal temperature of the vessel, and the final is the inlet/outlet temperatures of the heat exchanger.

Important equations:

a) Heat transfers calculations:

$$U = \frac{Q_{13}}{A(t_1 - t_3)}$$

where,

t_1 : steam temperature inside the jacket

t_3 : bulk average temperature of the fluid inside the tank ($^{\circ}\text{C}$)

Q_{13} : amount of heat transferred to the vessel from the steam (W)

U : overall heat transfer coefficient ($\frac{\text{J}}{\text{s.m}^2.\text{C}}$)

A : heat transfer surface area (m^2)

$$\begin{aligned} Q_{hx} &= mCp(\Delta T) = m_{hx}Cp(T_{ihx} - T_{ohx})_{hx} \\ Q_c &= mCp(\Delta T) = m_cCp(T_{oc} - T_{ic})_{c.c} \\ Q_{13} &= Q_{hx} + Q_c + Q_{surr} \end{aligned}$$

where,

$T_{o,hx}$: temperature going out of heat exchanger ($^{\circ}\text{C}$)

$T_{i,hx}$: temperature of fluid entering the heat exchanger ($^{\circ}\text{C}$)

$(T_{oc} - T_{ic})_{c.c}$: temperature rise of the cooling water in the coil ($^{\circ}\text{C}$)

Q_{hx} : amount of heat removed in the external heat exchanger (W)

Q_c : heat removed by the cooling coil (W)

Q_{surr} : heat loss to the surrounding, negligible (W)

Cp : heat capacity of fluid

m : flow rate ($\frac{\text{kg}}{\text{s}}$)

b) Nusselt number, Reynolds number and Prandtl number calculations:

Nusselt number, $Nu = 0.36 Re^{\frac{2}{3}} Pr^{\frac{1}{3}}$

Impeller Reynolds number, $Re_{imp} = D_I^2 n \frac{\rho}{\mu}$

Fluid Prandtl number, $Pr = Cp \frac{\mu}{k}$

Tank Nusselt number, $Nu_T = h \frac{D_T}{k}$

where,

D_I : impeller diameter

n : speed of rotation of the impeller, revolutions per second

ρ : density of the fluid

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D_T : tank diameter

μ : viscosity of the fluid

k : thermal conductivity of the fluid

Other calculations:

c) Heat Transfer Surface area, $A = 2\pi r * h$

where,

r: radius of vessel = 0.133 m

h: height of vessel (m)

d) Heat transfer coefficient, U:

$$h_t = \frac{1}{\frac{1}{U} + \frac{1}{h_0}}$$

where,

h_0 : heat transfer coefficient at outer wall ($\frac{J}{m^2 * s * K}$)

h_i : heat transfer coefficient at inner wall ($\frac{J}{m^2 * s * K}$)

e) Heat transfer coefficient in terms of physical properties and in terms of Nu, Re number:

$$\log(Nu_T / (Pr^{\frac{1}{3}})) = \log(0.36) + 0.67 \log(Re_{imp})$$

$$\frac{h \frac{D_T}{k}}{(Cp \frac{\mu}{k})^{.33} (\frac{\mu^{.14}}{\mu})} = 0.36 (D_T^2 n \frac{\rho}{\mu})^{0.67}$$

4. Results

For this experiment, the main goal was to measure the heat transfer coefficient inside the vessel and how this one would change with respect to the impeller speed, the varying temperature, and the presence of baffles in the system. Thus, the flow rate of steam into the vessel was kept constant for the first portion of the experiment at a low impeller speed of 600 RPM. Through the system, the water flows in the cooling coils at a temperature of about 65 degrees Celsius, which would let calculate the value of Q_{13} .

To obtain this value, the heat transfer rate from the steam to the vessel is calculated by considering an energy balance around the vessel. With this assumption, we obtain that Q_{13} is the sum of the heat removed from the heat exchanger (Q_{hx}), the heat removed from the coils (Q_c), and the heat loss to the surroundings (Q_{surr}). The Q_{surr} is assumed to be zero since the vessel is insulated.

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The calculated data for the first portion of the experiment is summarized in the following table:

Table 1 - Calculated and Literature Physical Properties of the Fluid for the Calculations of the Wall HTC for the Unbaffled Agitated Tank (as a function of impeller speed)

Trial	Q_{hx}	Q_{coils}	Q_{13}	Nusselt Number	Impeller Reynolds number	Prandtl number	h_i	U	h_o
Low Speed Impeller	$1062 \frac{J}{s}$	$2483 \frac{J}{s}$	$3545 \frac{J}{s}$	7,110	1,380,796	4.04	$12,569 \frac{W}{m^2 K}$	$416.5 \frac{W}{m^2 s}$	$430.8 \frac{W}{m^2 K}$
Middle Speed Impeller	$1058 \frac{J}{s}$	$2696 \frac{J}{s}$	$3755 \frac{J}{s}$	9,995	2,301,327	4.04	$17,669 \frac{W}{m^2 K}$	$300.1 \frac{W}{m^2 s}$	$305.3 \frac{W}{m^2 K}$
High Speed Impeller	$993 \frac{J}{s}$	$3239 \frac{J}{s}$	$4233 \frac{J}{s}$	11,598	2,876,658	4.04	$20,503 \frac{W}{m^2 K}$	$339.1 \frac{W}{m^2 s}$	$344.8 \frac{W}{m^2 K}$

As seen in table 1, Q_{13} for the unbaffled agitated tank has an average of $3,844 \frac{J}{s}$, which leads to the calculation of an average value of $16,914 \frac{W}{m^2 K}$ for h_i , an average value of $360.3 \frac{W}{m^2 K}$ for h_o , and an average value of $352 \frac{W}{m^2 s}$ for U.

When calculating the value of U in the other route, we can get an average value of $856 \frac{W}{m^2 s}$. The values for the other variables with this other route are summarized in table 2.

Table 2 - Calculating U as a function of impeller speed for Unbaffled Tank

Impeller Speed	$h_o (\frac{W}{m^2 K})$	$h_i (\frac{W}{m^2 K})$	U ($\frac{W}{m^2 s}$)
600	904.09	12567.66	843.42
1000	904.09	17667.16	860.08
1250	904.09	20500.62	865.90

The correlations for these numbers can be plotted by using the following relation:

$$\log\left(\frac{Nu_T}{Pr^{\frac{1}{3}}}\right) = \log(0.36) + 0.67 \log(Re)$$

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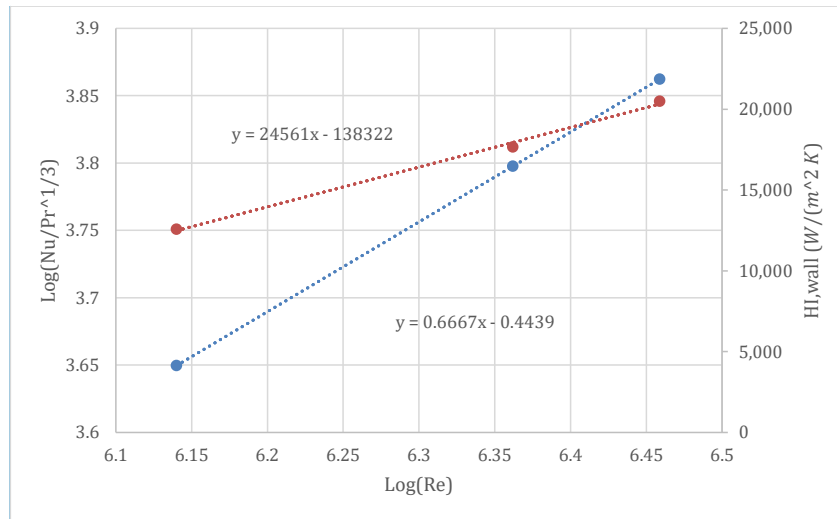


Figure 1 - Un baffled $\text{Log}(\text{Re})$ vs. $\text{Log}\left(\frac{\text{Nu}}{\text{Pr}^{1/3}}\right)$ at Different Impeller Speeds

In figure 1, the blue line represents the plot of the left-hand side of the equation versus the logarithm of the impeller Reynolds number. On the other hand, the red line represents the points for $h_{i,\text{wall}}$ as a function of the impeller Reynolds number. Just as expected, the logarithm plot gave a line with a slope of 0.67, displaying an increasing trend. Thus, this shows that as the impeller speed increases, so does the Reynolds number.

Similarly, the calculations for the un baffled agitated tank as a function of temperature were obtained by using the same correlations and assumptions. For this part of the experiment, the impeller speed was kept constant, though the flow rates for the heat exchanger were used to obtain two different vessel temperatures. The data for these calculations is summarized in tables 3 and 4 for both routes.

Table 3 – Calculated and Literature Physical Properties of the Fluid for the Calculations of the Wall HTC for the Un baffled Agitated Tank (as a function of temperature)

Trial	Q_{hx}	Q_{coils}	Q_{13}	Nusselt Number	Impeller Reynolds number	Prandtl number	h_i	U	h_o
Low Flowrate	$654 \frac{\text{J}}{\text{s}}$	$3997 \frac{\text{J}}{\text{s}}$	$4651 \frac{\text{J}}{\text{s}}$	9,995	2,301,327	4.041	$17,669 \frac{\text{W}}{\text{m}^2 \text{K}}$	$216 \frac{\text{W}}{\text{m}^2 \text{s}}$	$218 \frac{\text{W}}{\text{m}^2 \text{K}}$

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High Flowrate	$3032 \frac{J}{s}$	$3828 \frac{J}{s}$	$6860 \frac{J}{s}$	9,995	2,301,327	4.041	$17,669 \frac{W}{m^2 K}$	$390 \frac{W}{m^2 s}$	$399 \frac{W}{m^2}$	Commented [DL31]: Do these numbers keep the same?
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Table 4 - Calculating U as a function of temperature for Unbaffled Tank

$h_o (\frac{W}{m^2 K})$	$h_i (\frac{W}{m^2 K})$	$U (\frac{W}{m^2 s})$
490.61	17667.16	477.35
456.54	17667.16	445.04

For experiment B, the same procedure was used to calculate $h_{i,wall}$ as a function of impeller speed and temperature. Nevertheless, the vessel is operating under baffled condition. Thus, the following results were obtained.

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Table 5 – Calculated and Literature Physical Properties of the Fluid for the Calculations of the Wall HTC for the Baffled Agitated Tank (as a function of impeller speed)

Trial	Q_{hx}	Q_{coils}	Q_{13}	Nusselt Number	Impeller Reynolds number	Prandtl number	h_i	U	h_o
Low Speed Impeller	$2354 \frac{J}{s}$	$2816 \frac{J}{s}$	$5170 \frac{J}{s}$	7,110	1,380,796	4.041	$12,569 \frac{W}{m^2 K}$	$1,446 \frac{W}{m^2 s}$	$1633 \frac{W}{m^2 K}$
Middle Speed Impeller	$2179 \frac{J}{s}$	$3105 \frac{J}{s}$	$5285 \frac{J}{s}$	7,376	1,459,041	4.041	$13,040 \frac{W}{m^2 K}$	$1,940 \frac{W}{m^2 s}$	$2279 \frac{W}{m^2 K}$
High Speed Impeller	$1830 \frac{J}{s}$	$3461 \frac{J}{s}$	$5292 \frac{J}{s}$	9,966	2,291,431	4.041	$17,619 \frac{W}{m^2 K}$	$2,331 \frac{W}{m^2 s}$	$2686 \frac{W}{m^2 K}$

Table 6 - Calculating U as a function of impeller speed for Baffled Tank

Impeller Speed	$h_o (\frac{W}{m^2 K})$	$h_i (\frac{W}{m^2 K})$	$U (\frac{W}{m^2 s})$
600	714.43	12567.64	676.00
634	714.43	13037.82	677.32

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995.7	714.43	17615.90	686.58
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Based on these values, the average of the heat transfer coefficient for the baffled tank as a function of the impeller speed is equal to $1,349 \frac{W}{m^2 K}$.

Similarly, the correlations for these numbers can be plotted by using the logarithm equation into the following graph:

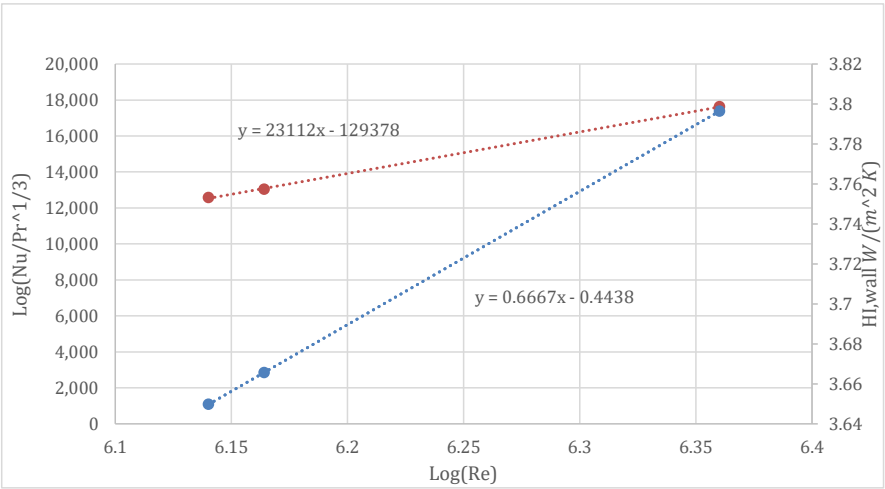


Figure 2 - Baffled Log (Re) vs. $\text{Log}\left(\frac{Nu_T}{Pr^{1/3}}\right)$ at Different Impeller Speeds

Just like in part A of the experiment, the logarithm relationship gave out the expected slope for the equation of the line, while the hi of the wall displayed an increasing trend for all its values as the Reynolds number increased.

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Finally, the values required for the calculation of the overall heat transfer coefficient as a function of temperature with the baffled condition, are summarized in the following tables by using both calculation methods.

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Table 7 – Calculated and Literature Physical Properties of the Fluid for the Calculations of the Wall HTC for the Baffled Agitated Tank (as a function of temperature)

Trial	Q_{hx}	Q_{coils}	Q_{13}	Nusselt Number	Impeller Reynolds number	Prandtl number	h_i	U	h_o
Low Flowrate	$1355 \frac{J}{s}$	$5094 \frac{J}{s}$	$6448 \frac{J}{s}$	8,505	1,806,541	4.041	$15,036 \frac{W}{m^2 K}$	$455 \frac{W}{m^2 s}$	$469 \frac{W}{m^2 K}$
High Flowrate	$1458 \frac{J}{s}$	$2926 \frac{J}{s}$	$4384 \frac{J}{s}$	8,505	1,806,541	4.041	$15,036 \frac{W}{m^2 K}$	$276 \frac{W}{m^2 s}$	$281 \frac{W}{m^2 K}$

Table 8 - Calculating U as a function of temperature for Baffled Tank

$h_o (\frac{W}{m^2 K})$	$h_i (\frac{W}{m^2 K})$	$U (\frac{W}{m^2 s})$
526.68	15033.44	508.85
445.36	15033.44	432.55

Based on these numbers, the average heat transfer coefficient as a function of temperature for the baffled tank is $418 \frac{W}{m^2 s}$.

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5. Discussion

In chemical engineering, the study of agitated tanks, also known as CSTR, is of high interest to determine how mixing can affect the heat transfer coefficients through different fluids that are part of the system. For this experiment, the heat transfer coefficient was investigated as a function of the impeller speed, varying temperatures, and the presence of baffles. To perform these calculations, it was necessary to obtain the values for the dimensionless numbers that correlate the heat transfer coefficient to the performing conditions of the tank (diameter, impeller speed, flowrates, etc.).

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Based on this, it can be observed that as the impeller speed increases, so does the Reynolds number. This happens because the Reynolds number is a way to measure turbulence in a fluid, which shows how mixing is directly proportional to the impeller speed. These results can be seen in figures 1 and 2 in the result section of the lab report.

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Additionally, it was found that the overall heat transfer coefficient increases as the impeller speed increases because when the fluid is mixed, the differential temperature of what is exchanging is greater. Thus, the heat transfer increases as well. Furthermore, when comparing the system with baffled and unbaffled conditions, it was found that the overall heat transfer coefficient increases with the presence of baffles. This is because baffles increase mixing, which ultimately increases the heat transfer between the two surfaces.

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Two different methods were used to obtain the overall heat transfer coefficient. It is believed that the first method which involves the calculation of the heat transfer to vessel from the steam and using the energy balance across the agitated vessel is more accurate because the h_o values are not dependent on the flowrate of the incoming steam.

Finally, the agreement between the correlated values and the measured values was not as accurate as expected because the impeller used to agitate the fluid does not mix the fluid homogeneously throughout the process. This inconsistency is shown in figures 1 and 2, in which the graph that relates the heat transfer coefficient of the wall did increase as the Reynolds number increased but it did not match with the expected slope for the equation of the line.

6. References

[1] Ulku, S., and T. Cakaloz. "Heat Transfer in Vessels Agitated With Propeller." *Proceeding of International Heat Transfer Conference 8*, 1986, doi:10.1615/ihtc8.4400. Accessed 5 Feb. 2020

[2] EPA, Environmental Protection Agency, nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=9101ILLQ.TXT. Accessed 5 Feb. 2020

[3] R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena*, 2nd ed. John Wiley and Sons, Inc., 2002.

[4] R. H. Perry and D. W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7th ed. McGraw-Hill, 1997.

[5] "Water - Dynamic and Kinematic Viscosity" The Engineering Toolbox, www.engineeringtoolbox.com/water-liquid-gas-thermal-conductivity-temperature-pressure-d_2012.html. Accessed 5 Feb. 2020

[6] Perlman, Howard, and USGS. "Water Density." *The USGS Water Science School*, water.usgs.gov/edu/density.html. Accessed 5 Feb. 2020

7. Appendix I: Data Tabulation/Graphs

Raw Data for Part A

Table 9a – Raw Data for the Calculations of the Wall HTC for the Unbaffled Agitated Tank (as a function of impeller speed)

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Heat Transfer to a Fluid in a Stirred Tank

University of Illinois at Chicago

Trial	Impeller Speed (rpm)	Temp of the Cooling Water in (°C)	Temp of the Cooling Water Out (°C)	Flow Rate of the Heat Exchanger (GPM)	Temp of Fluid Inside the Tank (°C)	Temp of Fluid Outside the Tank (°C)	Temp of Steam In (°C)	Temp of Steam Out (°C)
Low Speed Impeller	600	3.89	16.7	0.55	58	66	73	6.11
Middle Speed Impeller	1000	3.89	17.8	0.88	45	63	67	5.45
High Speed Impeller	1250	3.89	20.6	1.10	42	65	64	4.44

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Table 9b – Raw Data for the Calculations of the Wall HTC for the Unbaffled Agitated Tank (as a function of impeller speed)

Trial	Impeller Speed (rpm)	Temp of Heat Exchanger in (°C)	Temp of Heat Exchanger out (°C)	Flowrate of Steam	Flowrate of Water
Low Speed Impeller	600	68	43.3	$2.78 \frac{kg}{min}$	$0.616 \frac{kg}{min}$
Middle Speed Impeller	1000	59	34.4	$2.78 \frac{kg}{min}$	$0.616 \frac{kg}{min}$
High Speed Impeller	1250	62	38.89	$2.78 \frac{kg}{min}$	$0.616 \frac{kg}{min}$

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Table 10 – Calculated and Literature Physical Properties of the Fluid for the Calculations of the Wall HTC for the Unbaffled Agitated Tank (as a function of impeller speed)

Trial	Q_{hx}	Q_{coils}	Q_{13}	Nusselt Number	Impeller Reynolds number	Prandtl number	h_i	U	h_o
Low Speed Impeller	$1062 \frac{J}{s}$	$2483 \frac{J}{s}$	$3545 \frac{J}{s}$	7,110	1,380,796	4.041	$12,569 \frac{W}{m^2 K}$	$416.5 \frac{W}{m^2 s}$	$430.8 \frac{W}{m^2 K}$
Middle Speed Impeller	$1058 \frac{J}{s}$	$2696 \frac{J}{s}$	$3755 \frac{J}{s}$	9,995	2,301,327	4.041	$17,669 \frac{W}{m^2 K}$	$300.1 \frac{W}{m^2 s}$	$305.3 \frac{W}{m^2 K}$

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High Speed Impeller	$993 \frac{J}{s}$	$3239 \frac{J}{s}$	$4233 \frac{J}{s}$	11,598	2,876,658	4.041	$20,503 \frac{W}{m^2 K}$	$339.1 \frac{W}{m^2 s}$	$344.8 \frac{W}{m^2 K}$
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Table 11a – Raw Data for the Calculations of the Wall HTC for the Unbaffled Agitated Tank (as a function of temperature)

Trial	Impeller Speed (rpm)	Temp of the Cooling Water in (°C)	Temp of the Cooling Water Out (°C)	Temp of Fluid Inside the Tank (°C)	Temp of Fluid Outside the Tank (°C)	Temp of Steam In (°C)	Temp of Steam Out (°C)
Low Flowrate	1000	4.55	19.4	41	62	79	7.35
High Flowrate	1000	5.34	16.8	44	69	75	6.85

Table 11b – Raw Data for the Calculations of the Wall HTC for the Unbaffled Agitated Tank (as a function of temperature)

Trial	Impeller Speed (rpm)	Flowrate of Steam	Flowrate of Water	Temp of Heat Exchanger in (°C)	Temp of Heat Exchanger out (°C)
Low Flowrate	1000	$1.13 \frac{kg}{min}$	$3.86 \frac{kg}{min}$	58	49.7
High Flowrate	1000	$1.85 \frac{kg}{min}$	$4.79 \frac{kg}{min}$	65	41.5

Table 12 – Calculated and Literature Physical Properties of the Fluid for the Calculations of the Wall HTC for the Unbaffled Agitated Tank (as a function of temperature)

Trial	Q_{hx}	Q_{coils}	Q_{13}	Nusselt Number	Impeller Reynolds number	Prandtl number	h_i	U	h_o
Low Flowrate	$654 \frac{J}{s}$	$3997 \frac{J}{s}$	$4651 \frac{J}{s}$	9,995	2,301,327	4.041	$17,669 \frac{W}{m^2 K}$	$216 \frac{W}{m^2 s}$	$218 \frac{W}{m^2 K}$
High Flowrate	$3032 \frac{J}{s}$	$3828 \frac{J}{s}$	$6860 \frac{J}{s}$	9,995	2,301,327	4.041	$17,669 \frac{W}{m^2 K}$	$390 \frac{W}{m^2 s}$	$399 \frac{W}{m^2 K}$

Raw Data for Part B**Table 13a – Raw Data for the Calculations of the Wall HTC for the Baffled Agitated Tank
(as a function of impeller speed)**

Trial	Impeller Speed (rpm)	Temp of the Cooling Water in (°C)	Temp of the Cooling Water Out (°C)	Flow Rate of the Heat Exchanger (GPM)	Temp of Fluid Inside the Tank (°C)	Temp of Fluid Outside the Tank (°C)	Temp of Steam In (°C)	Temp of Steam Out (°C)
Low Speed Impeller	600	4.44	17.1	0.55	77.7	78.0	84.0	6.67
Middle Speed Impeller	634	4.44	18.4	0.88	82.2	82.0	87.0	7.77
High Speed Impeller	995.7	4.44	20.0	1.10	85.0	90.0	89.0	8.33

**Table 13b – Raw Data for the Calculations of the Wall HTC for the Baffled Agitated Tank
(as a function of impeller speed)**

Trial	Impeller Speed (rpm)	Temp of Heat Exchanger in (°C)	Temp of Heat Exchanger out (°C)	Flowrate of hot water	Flowrate of cold water
Low Speed Impeller	600	76	49	$1.25 \frac{kg}{min}$	$3.19 \frac{kg}{min}$
Middle Speed Impeller	634	71	46	$1.25 \frac{kg}{min}$	$3.19 \frac{kg}{min}$
High Speed Impeller	995.7	72	51	$1.25 \frac{kg}{min}$	$3.19 \frac{kg}{min}$

Table 14 – Calculated and Literature Physical Properties of the Fluid for the Calculations of the Wall HTC for the Baffled Agitated Tank (as a function of impeller speed)

Heat Transfer to a Fluid in a Stirred Tank

University of Illinois at Chicago

Trial	Q_{hx}	Q_{coils}	Q_{13}	Nusselt Number	Impeller Reynolds number	Prandtl number	h_i	U	h_o
Low Speed Impeller	$2354 \frac{J}{s}$	$2816 \frac{J}{s}$	$5170 \frac{J}{s}$	7,110	1,380,796	4.041	$12,569 \frac{W}{m^2 K}$	$1,446 \frac{W}{m^2 s}$	$1633 \frac{W}{m^2 K}$
Middle Speed Impeller	$2179 \frac{J}{s}$	$3105 \frac{J}{s}$	$5285 \frac{J}{s}$	7,376	1,459,041	4.041	$13,040 \frac{W}{m^2 K}$	$1,940 \frac{W}{m^2 s}$	$2279 \frac{W}{m^2 K}$
High Speed Impeller	$1830 \frac{J}{s}$	$3461 \frac{J}{s}$	$5292 \frac{J}{s}$	9,966	2,291,431	4.041	$17,619 \frac{W}{m^2 K}$	$2,331 \frac{W}{m^2 s}$	$2686 \frac{W}{m^2 K}$

Table 15a – Raw Data for the Calculations of the Wall HTC for the Baffled Agitated Tank (as a function of temperature)

Trial	Impeller Speed (rpm)	Temp of the Cooling Water in (°C)	Temp of the Cooling Water Out (°C)	Temp of Fluid Inside the Tank (°C)	Temp of Fluid Outside the Tank (°C)	Temp of Steam In (°C)	Temp of Steam Out (°C)
Low Flowrate	785	2.35	17.6	49	67	74	6.45
High Flowrate	785	3.64	12.4	43	65	71	6.12

Table 15b – Raw Data for the Calculations of the Wall HTC for the Baffled Agitated Tank (as a function of temperature)

Trial	Impeller Speed (rpm)	Flowrate of Steam	Flowrate of Water	Temp of Heat Exchanger in (°C)	Temp of Heat Exchanger out (°C)
Low Flowrate	785	$2.14 \frac{kg}{min}$	$3.12 \frac{kg}{min}$	62	51.5
High Flowrate	785	$2.96 \frac{kg}{min}$	$5.16 \frac{kg}{min}$	68	56.7

Table 16 – Calculated and Literature Physical Properties of the Fluid for the Calculations of the Wall HTC for the Baffled Agitated Tank (as a function of temperature)

Trial	Q_{hx}	Q_{coils}	Q_{13}	Nusselt Number	Impeller Reynolds number	Prandtl number	h_i	U	h_o
Low Flowrate	$1355 \frac{J}{s}$	$5094 \frac{J}{s}$	$6448 \frac{J}{s}$	8,505	1,806,541	4.041	$15,036 \frac{W}{m^2 K}$	$455 \frac{W}{m^2 s}$	$469 \frac{W}{m^2 K}$
High Flowrate	$1458 \frac{J}{s}$	$2926 \frac{J}{s}$	$4384 \frac{J}{s}$	8,505	1,806,541	4.041	$15,036 \frac{W}{m^2 K}$	$276 \frac{W}{m^2 s}$	$281 \frac{W}{m^2 K}$

Table 17 - Calculating U as a function of impeller speed for Unbaffled Tank

Impeller Speed	$h_o (\frac{W}{m^2 K})$	$h_i (\frac{W}{m^2 K})$	$U (\frac{W}{m^2 s})$
600	904.09	12567.66	843.42
1000	904.09	17667.16	860.08
1250	904.09	20500.62	865.90

Table 18 - Calculating U as a function of temperature for Unbaffled Tank

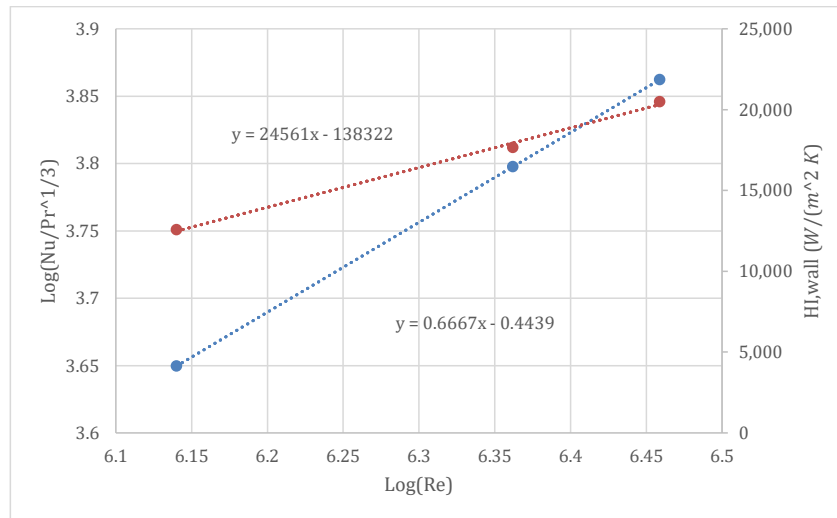
Temperature of Heat Exchanger In (°C)	Temperature of Heat Exchanger Out (°C)	$h_o (\frac{W}{m^2 K})$	$h_i (\frac{W}{m^2 K})$	$U (\frac{W}{m^2 s})$
58	49.7	490.61	17667.16	477.35
65	41.5	456.54	17667.16	445.04

Table 19 - Calculating U as a function of impeller speed for Baffled Tank

Impeller Speed	$h_o (\frac{W}{m^2 K})$	$h_i (\frac{W}{m^2 K})$	$U (\frac{W}{m^2 s})$
600	714.43	12567.64	676.00
634	714.43	13037.82	677.32
995.7	714.43	17615.90	686.58

Table 20 - Calculating U as a function of temperature for Baffled Tank

Temperature of Heat Exchanger In (°C)	Temperature of Heat Exchanger Out (°C)	$h_o \left(\frac{W}{m^2 K} \right)$	$h_i \left(\frac{W}{m^2 K} \right)$	$U \left(\frac{W}{m^2 s} \right)$
62	51.5	526.68	15033.44	508.85
68	56.7	445.36	15033.44	432.55

Figure 3 - Un baffled $\text{Log}(\text{Re})$ vs. $\text{Log}\left(\frac{Nu}{Pr^{1/3}}\right)$ at Different Impeller Speeds

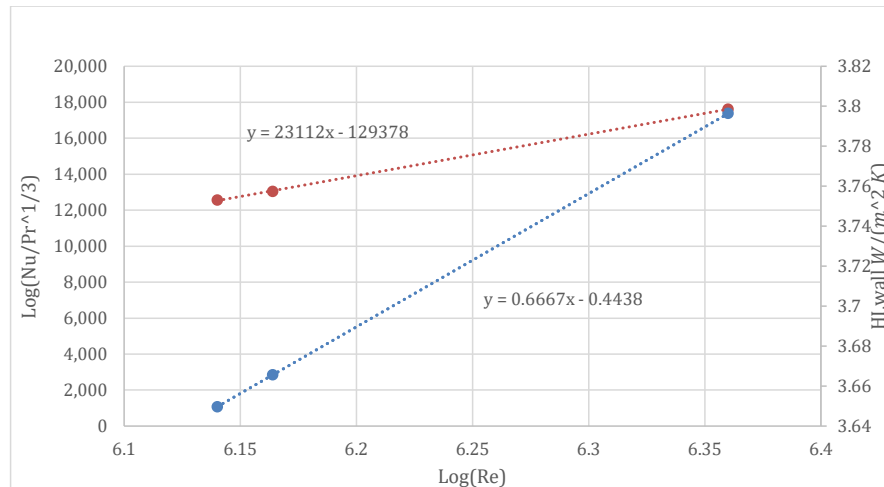


Figure 4 - Baffled Log (Re) vs. $\text{Log} \left(\frac{Nu_T}{Pr^{\frac{1}{3}}} \right)$ at Different Impeller Speeds

8. Appendix II: Error Analysis

After a thorough examination of the continuous stirred tank, we noticed possible sources of error resulting from the equipment and experimenters. During the unbaffled portion of the experiment, we noticed that the tank was unable to exceed 45°C. We credit the lower temperature to a fault in the steam system within the laboratory, being unable to provide enough steam to transfer heat to the tank conductively. The ineffective heat transfer did not allow us to perform the experiments at similar temperature ranges; however, after accessing the data and accounting for the difference in temperature and flow rates, the dependent values such as heat transfer and dimensionless numbers are reasonable.

The complexity of the apparatus adds to the possibility of errors resulting from operating inconsistencies. For example, the flow rate of steam was measured multiple times during the experiment for both baffled and unbaffled arrangements. Still, there may have been times where fluctuations of steam flow rate may have existed. Additionally, water temperature is measured at the top of the tank. An extra thermometer at the bottom of the tank may expose a temperature gradient within the container. Finally, near our defined high impeller speed we heard an audible clatter from within the tank this was later discovered to be the baffles drawn toward the cooling coil due to the flow within the tank. The baffles lifting from the wall of the vessel may have affected the fluid dynamics and possibly contributed to potentially hazardous equipment operation.

9. Appendix III: Sample Calculation

Commented [DL48]: 8/10

Quantitative analysis is needed. Like the comparison of the values of the dimensionless groups calculated from the empirical equation and obtained from the experiment.

Commented [DL49]: 8/10

Fatal calculation errors.

Specify what trial you're presenting.

The flow rate of the cold water can be calculated as follows in units of kg/min.

Mass of dry cylinder = 103 g

Mass of cylinder with water = 357.2 g

Time taken for collection of water = 6 seconds

$$M_c = \frac{(357.2 \text{ g} - 103 \text{ g})}{6 \text{ seconds}} \times \frac{1 \text{ kg}}{10^3 \text{ g}} \times \frac{60 \text{ seconds}}{1 \text{ minute}} = 2.542 \frac{\text{kg}}{\text{min}}$$

As for the calculation of the flow rate of hot water, it can be obtained similarly:

$$M_c = \frac{(205.5 \text{ g} - 102.7 \text{ g})}{10 \text{ seconds}} \times \frac{1 \text{ kg}}{10^3 \text{ g}} \times \frac{60 \text{ seconds}}{1 \text{ minute}} = 0.6168 \frac{\text{kg}}{\text{min}}$$

Next, we can calculate the heat transfer removed from the heat exchanger in units of J/s by using the following relation:

$$Q_{hx} = m_{hx} C_p (T_{hx,hot} - T_{hx,cold})$$

$$Q_{hx} = \left(0.6168 \frac{\text{kg}}{\text{minute}} \times \frac{1 \text{ minute}}{60 \text{ s}} \right) \left(4184 \frac{\text{J}}{\text{kg} \cdot \text{K}} \right) (68^\circ\text{C} - 43.3^\circ\text{C}) = 1062 \frac{\text{J}}{\text{s}}$$

As well as calculating the heat transfer removed by the cooling coils by using a similar relation:

$$Q_c = m_c C_p (T_{c,hot} - T_{c,cold})$$

$$Q_c = \left(2.78 \frac{\text{kg}}{\text{min}} \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ min}}{60 \text{ s}} \right) \left(4184 \frac{\text{J}}{\text{kg} \cdot \text{K}} \right) (16.7^\circ\text{C} - 3.89^\circ\text{C}) = 2483 \text{ J/s}$$

Finally, the total heat transfer to the system is equal to the following:

$$Q_{13} = Q_{hx} + Q_c + Q_{surr}$$

For this step, we assume that the Q_{surr} which is the heat loss from the recirculation pipes and due to evaporation of water from the vessel is equal to zero because the vessel is insulated so no heat is lost to the surroundings.

$$Q_{13} = 1062 + 2483 = 3545 \frac{\text{J}}{\text{s}}$$

Which then lets us calculate the overall heat transfer coefficient with the following relation:

$$U_{wall} = \frac{Q_{13}}{A(T_{steam} - T_{vessel})}$$

In which A is given by the following:

$$A = 2\pi r h$$

Which can be simplified to the following expression:

$$A = \pi D h$$

Where D represents the diameter of the vessel and h represents the length or height

$$A = (\pi)(0.3556 \text{ m})(0.508 \text{ m})$$

$$A = 0.5675 \text{ m}^2$$

$$U_{wall} = \frac{3545 \frac{J}{s}}{(0.5675 \text{ m}^2)(73^\circ\text{C} - 58^\circ\text{C})} = \frac{416.5 \text{ W}}{\text{m}^2\text{K}}$$

Commented [DL51]: This is the right unit.

The inner wall heat transfer coefficient in an unbaffled vessel containing a Newtonian fluid is given by the following relation:

$$\frac{h_i D_T}{k} = (0.36) \left(\frac{D_I^2 n \rho}{\mu} \right)^{\frac{2}{3}} \left(\frac{Cp \mu}{k} \right)^{\frac{1}{3}}$$

$$\frac{h D_T}{k} = Nu_T$$

$$\frac{D_I^2 n \rho}{\mu} = Re_{imp}$$

$$\frac{Cp \mu}{k} = Pr$$

Thus, the calculations will go as follows:

$$\frac{(0.0889 \text{ m}^2)(10 \text{ rps}) \left(997 \frac{\text{kg}}{\text{m}^3} \right)}{(0.0006419 \text{ Pa} \cdot \text{s})} = Re_{imp}$$

$$1,380,796 = Re_{imp}$$

$$\frac{(0.0006419 \text{ Pa} \cdot \text{s}) \left(4184 \frac{\text{J}}{\text{kg} \cdot \text{K}} \right)}{\left(0.62856 \frac{\text{W}}{\text{m} \cdot \text{K}} \right)} = Pr$$

$$Pr = 4.041$$

$$Nu = 0.36 (Re_{imp})^{\frac{2}{3}} (Pr)^{\frac{1}{3}}$$

$$Nu = 0.36 (1,380,796)^{\frac{2}{3}} (4.041)^{\frac{1}{3}} = 7,110$$

Commented [DL52]: The square should be outside the parenthesis.

Therefore, we can solve for h_i as follows:

$$h_i = (0.36) \left(\frac{D_I^2 n \rho}{\mu} \right)^{\frac{2}{3}} \left(\frac{Cp \mu}{k} \right)^{\frac{1}{3}} \frac{k}{D_T}$$

$$h_i = \frac{Nu * k}{D_T}$$

$$h_i = \frac{(7,110) * \left(0.62856 \frac{\text{W}}{\text{m} \cdot \text{K}} \right)}{(0.3556 \text{ m})}$$

$$h_i = 12,567 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

When plugging in our numbers, we find the value of h_i to be equal to $12,567 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$

Finally, h_o can be calculated by using the following equation:

$$U_i = \frac{1}{\frac{1}{h_i} + \frac{1}{h_o}}$$

$$416.5 = \frac{1}{\frac{1}{12,567} + \frac{1}{h_o}}$$

$$h_o = 430.78 \frac{W}{m^2 K}$$

On the other hand, the calculation of h_o can also be obtained by using the following route:

Calculating h_o for unbaffled tank as a function of impeller speed:

$$\Gamma = \frac{m_{st}}{\pi D_{T,o}}$$

where m_{st} = the mass rate of steam condensation

$D_{T,o}$ = the outer diameter of the vessel wall

Γ = mass rate of steam condensation per wetted perimeter

$$\Gamma = \frac{10.28 g/s}{\pi(0.3556m)} = 9.20198 \frac{g}{m \cdot s} = 0.009202 \frac{kg}{m \cdot s}$$

$$\frac{h_o L_T}{k} = 0.9255 \left(\frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{1/3}$$

$$h_o = 0.9255 \left(\frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{1/3} k \times \frac{1}{L_T}$$

$$h_o = 0.9255 \left(\frac{(0.508m)^3 (997 \frac{kg}{m^3})^2 (9.80665 \frac{m}{s^2})}{(0.2822m \cdot Pa \cdot s) (0.009202 \frac{kg}{m \cdot s})} \right)^{1/3} \cdot \left(0.62856 \frac{W}{m \cdot K} \right) \frac{1}{0.508m} = 904.09 \frac{W}{m^2 K}$$

Calculating h_i for unbaffled tank as a function of impeller speed:

$$h_i = Nu_T \times \frac{k}{D_T}$$

$$h_i = 7110 \times \frac{0.62856 \frac{W}{m \cdot K}}{0.3556m}$$

$$h_i = 12567.66 \frac{W}{m^2 K}$$

10. Individual Team Contributions

Name: Javier Eduardo Mendieta	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Record data/ Measured
Pre-Lab Editing	2	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	1	Contributed
Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	2	Contributed
Project Safety Evaluation (prelab)	2	Contributed
Final Lab Editing	2	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	3	Contributed
Theory (Final Lab)	3	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Contributed
Conclusion (final lab)	2	Contributed

References (final and/or prelab)	1	Contributed
Data Tabulation/Graphs (final)	2	Completed
Error Analysis (final lab)	1	Contributed
Sample Calculations (final Lab)	3	Completed
PowerPoint Presentation	1	Contributed
Total	37	

Name: Omar Zahdan	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	2	Contributed
Experimental Objective (Prelab)	2	Contributed
Apparatus (prelab)	2	Contributed
Materials & Supplies (prelab)	2	Contributed
Procedure (prelab)	2	Completed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	3	Contributed
Executive Summary (final lab)	2	Contributed

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Introduction (Final lab)	2	Contributed
Theory (Final Lab)	1	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Contributed
Conclusion (final lab)	2	Contributed
References (final and/or prelab)	1	Contributed
Data Tabulation/Graphs (final)	1	Contributed
Error Analysis (final lab)	2	Completed
Sample Calculations (final Lab)	2	Contributed
PowerPoint Presentation	1	Contributed
Total	34	

Name: Omar Takrouri	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	3	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	2	Contributed

Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	1	Contributed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	1	Contributed
Executive Summary (final lab)	1	Contributed
Introduction (Final lab)	2	Completed
Theory (Final Lab)	1	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Completed
Conclusion (final lab)	2	Contributed
References (final and/or prelab)	3	Contributed
Data Tabulation/Graphs (final)	2	Contributed
Error Analysis (final lab)	1	Contributed
Sample Calculations (final Lab)	1	Contributed
PowerPoint Presentation	2	Contributed
Total	31	

Name: Joel Finlon	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	1	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	1	Contributed
Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	2	Contributed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	4	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	2	Contributed
Theory (Final Lab)	2	Completed
Results (final lab)	1	Contributed
Discussion (final lab)	1	Contributed
Conclusion (final lab)	2	Completed
References (final and/or prelab)	3	Contributed

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Data Tabulation/Graphs (final)	1	Contributed
Error Analysis (final lab)	2	Contributed
Sample Calculations (final Lab)	1	Contributed
PowerPoint Presentation	1	Contributed
Total	31	

Name: Yusuf Sorunke	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	1	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	1	Contributed
Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	2	Contributed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	4	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	2	Contributed

Heat Transfer to a Fluid in a Stirred Tank**University of Illinois at Chicago**

Theory (Final Lab)	2	Completed
Results (final lab)	1	Contributed
Discussion (final lab)	1	Contributed
Conclusion (final lab)	2	Completed
References (final and/or prelab)	2	Contributed
Data Tabulation/Graphs (final)	1	Contributed
Error Analysis (final lab)	2	Contributed
Sample Calculations (final Lab)	1	Contributed
PowerPoint Presentation	1	Contributed
Total	30	